Ibrahim Baabdullah

Nursing and Health Informatics

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github.com/ibrahimbaabdullah-jpg | LinkedIn

PROFILE

Nursing professional and health informatics researcher with experience in clinical care, electronic health records, quantitative research and digital health projects. Holds a Master of Advanced Practice Nursing, with a dissertation awarded Distinction. Research interests include health information system usability, data quality, interoperability and clinical AI evaluation.

EDUCATION

Queensland University of Technology | Australia

Master of Advanced Practice Nursing — Jul 2022–Nov 2025

GPA: 5.80/7.00 | Dissertation: 6/7, Distinction | Supervisor: Dr Aaron Conway

Dissertation: Nursing Informatics Competence and Attitudes Towards Computers Among Final-Year Baccalaureate Nursing Students at Najran University: A Cross-Sectional Study in Saudi Arabia.

Relevant coursework: Epidemiology, Health Statistics, Qualitative Research, Leadership in Nursing, and Principles of Education for Health Professionals.

University of Newcastle | Australia

Bachelor of Nursing — Mar 2019–Dec 2021

GPA: 5.60/7.00 | 2021 College Commendation List for outstanding academic achievement

Relevant coursework: Knowledge Translation in Health Care; Leading, Teaching, Mentoring and Clinical Supervision for Health Professionals; Advanced Human Bioscience.

RESEARCH EXPERIENCE

Graduate Researcher — Jul 2023–Nov 2025

Queensland University of Technology | Brisbane, Australia

- Conceived, designed and conducted a quantitative cross-sectional study of self-reported nursing informatics competence and attitudes towards computer use among final-year nursing students at Najran University.
- Managed the research process, including proposal and protocol development, literature review, permissions to use the questionnaires, ethics approvals at QUT and Najran University, and coordination of data collection.
- Designed and administered a 60-item bilingual Qualtrics survey; managed responses from 48 students and performed data cleaning, coding, scoring and statistical analyses using IBM SPSS.
- Contributed methodological insights into assessing and interpreting nursing students' attitudes towards computer use in healthcare.
- Examined the relationship between self-reported informatics competence and attitudes towards computer use, identifying priorities for future research.
- Used Benner's Novice to Expert model as the primary framework, alongside perspectives on technology acceptance, motivation and self-assessment, to interpret the findings.
- Wrote a 169-page master's dissertation awarded Distinction and prepared an evidence-based report for Najran University, with recommendations for nursing informatics education, assessment practices and digital health readiness.

RESEARCH AND EDUCATIONAL OUTPUTS

Manuscript in preparation

Baabdullah, I. K. (2026). Nursing informatics competence and attitudes towards computers among final-year baccalaureate nursing students. Manuscript based on the QUT master's dissertation.

Educational resources

Baabdullah, I. K. (2025). Handwashing compliance in healthcare [Infographic]. Zenodo. doi.org/10.5281/zenodo.17823755

Baabdullah, I. K. (2025). New Graduate Nurses' Practical Guide to Clinical Leadership [Pamphlet]. Zenodo. doi.org/10.5281/zenodo.17823640

CLINICAL EXPERIENCE

Registered Nurse Mar 2023–Jul 2026

Najran University Hospital | Najran, Saudi Arabia

- Assessed patients' nursing needs, monitored vital signs and clinical progress, and communicated changes to the healthcare team.
- Administered prescribed medications and treatments, monitored patients' responses, and followed medication safety procedures.
- Documented assessments, interventions, and patient progress in the hospital's electronic health record system and communicated patient status, outstanding care needs, and safety concerns during shift handovers.
- Assisted with urgent assessment and emergency interventions in collaboration with the clinical team.
- Worked with physicians, nurses, and allied health professionals to coordinate care, communicate patient needs, and support admissions, transfers, and discharge preparation.
- Supported nursing students and newly appointed nurses through clinical guidance and orientation.

Student Nurse Jul 2019–Nov 2021

University of Newcastle | Newcastle, Australia

- Assisted patients with personal hygiene, oral care, toileting, repositioning, and pressure area care.
- Collected patient histories, measured vital signs, assessed pain, and reported findings to the supervising registered nurse (RN).
- Performed wound care, dressing changes, urinary catheterisation, and electrocardiogram (ECG) recordings under direct supervision.
- Supported the clinical team during electroconvulsive therapy (ECT) in theatre and during emergency responses, including Code Blue events.
- Reinforced patient education and explained care instructions under RN guidance.
- Administered prescribed subcutaneous and intramuscular injections, intravenous medications, and IV fluids under direct RN supervision.

RESEARCH AND TECHNICAL SKILLS

Research and analysis: Quantitative research design, literature review, critical appraisal, surveys, data cleaning, statistical analysis, IBM SPSS and Qualtrics.

Health informatics: EHR workflows, clinical documentation, data quality, HL7 FHIR R4, LOINC, UCUM, medication reconciliation and technology acceptance.

Project technologies: Python, SQL, pandas, NumPy, scikit-learn, PyTorch, XGBoost, Streamlit, FastAPI, SQLite, GitHub, pytest and GitHub Actions.

Additional tools: Power BI, Jupyter Notebook, EndNote, NVivo and REDCap.

Languages: Arabic and English.

HEALTH INFORMATICS PROJECTS

Ongoing research, educational and prototype development.

EHR Usability Lab **Website**

Built an EHR usability website comparing three interfaces through 50 tasks across eight nursing chart sections. Uses HTML, CSS and JavaScript with synthetic FHIR-format data, automatic timing and answer checks, browser storage, descriptive comparisons and CSV/HTML exports.

CareTrace Lite **Website**

Developed a clinical data-quality workbench for synthetic records, with record-level review and HL7 FHIR R4 Patient and Observation exports. Built with Python, Streamlit, pandas and SQLite, using LOINC, UCUM, source-to-export traceability and pytest.

Clinical AI Reliability Lab **Website (private)**

Developed a research workbench assessing ICU sepsis-model reliability across hospitals and under data perturbations, with a retrospective benchmark of 40,336 ICU records. Used Python, scikit-learn, XGBoost, PyTorch and FastAPI to examine calibration, alert burden and robustness.

MedRec Review **Website (private)**

Created a bilingual research prototype comparing fictional home and discharge medication lists, highlighting discrepancies and recording human decisions. Uses Python, Streamlit, Pydantic, pandas and Groq with openai/gpt-oss-120b, alongside deterministic comparison, limited dataset-based interaction checks and audit reports.

Nursing SBAR Handoff Summarizer **Website**

Built an Arabic-English educational application converting fictional nursing notes into Situation, Background, Assessment and Recommendation reports and flagging missing information. Uses Python, Streamlit and the Gemini API, with source-quotation checks, text and print exports, and automated tests.

MeasureTrace

Developed a research and educational workbench that traces how data-quality findings and evidence-supported corrections affect a synthetic nursing-assessment documentation measure. Built with Python, Streamlit, pandas and SQLite, with rule-based checks, review audit trails, recalculation, change analysis and reproducible experiment export and replay.

CareAccess ML **Website**

Built an experimental appointment no-show modelling and outreach-planning workbench using Python, Streamlit, scikit-learn and PyTorch, with saved-model inference and capacity-limited review lists. Implemented chronological evaluation, calibration and subgroup analysis, explicitly reporting model limitations.

VN Load **Website**

Created a simulation workbench comparing bedside-only, virtual-support and in-person-support admission and discharge workflows. Used Python, SimPy, Streamlit, pandas and Plotly to examine task allocation, waiting time and resource use through synthetic scenarios and sensitivity analyses.

NurseRecon **In development**

Built an initial prototype that compares explicit oral-temperature statements in synthetic nursing notes with FHIR R4 Observations and presents discrepancies for human review. Uses Python, FastAPI, React, TypeScript, SQLite and SQLAlchemy to preserve source evidence, review history and audit records.

TrialLens Clinical Trial Prescreening **Private website**

Built a bilingual research prototype for reviewing synthetic cases against trial eligibility criteria, with source-linked evidence, missing-information flags and human review records. Uses React, JavaScript, Vinext and web APIs, with Groq/Gemini options, ClinicalTrials.gov search and JSON/CSV exports.

ADDITIONAL DIGITAL PROJECTS

Ongoing development.

Arabic Career Interest Assessment **Website**

Built an exploratory Arabic career-interest assessment with 60 activities across six domains and O*NET-based occupation matching. Uses React, JavaScript, CSS and Recharts for scoring and results, with browser-local progress saving and explicit handling of tied scores.

SELECTED PROFESSIONAL DEVELOPMENT

- The Data Science of Health Informatics | Johns Hopkins University
- Python Basics | University of Michigan
- Databases and SQL for Data Science with Python | IBM
- Data Fundamentals | IBM
- BI Dashboards with Power BI | Codecademy
- Code Foundations Skill Path | Codecademy

HONOURS AND AWARDS

- 2026 | Recognised by the Saudi Arabian Cultural Mission for contributions to Saudi student activities in the USA.
- 2026 | Selected for a US$10,000-per-year Merit Scholarship Award, Boston University in the USA.
- 2022 | Awarded fully funded Ministry of Education scholarships for master's nursing study in Australia.
- 2021 | Recognised by the Saudi Arabian Cultural Mission for contributions to Saudi student activities in Australia.
- 2018 | Awarded fully funded Ministry of Education scholarships for bachelor's nursing study in Australia.
- 2016 | Recognised by the Saudi Arabian Cultural Mission for contributions to Saudi student activities in Canada.

VOLUNTEER EXPERIENCE

Volunteer experience with the Saudi Students Clubs in the United States, Australia and Canada; the Saudi Arabian Cultural Mission; Najran University; King Fahd University of Petroleum and Minerals; and the Neighborhood Centers Association in Jeddah, Saudi Arabia.

ADDITIONAL EDUCATION

Newcastle International College | Australia

Foundation Program (Health Studies) | Jun 2018–Mar 2019

Relevant coursework and achievements: Information Technology and Biology (High Distinction); Human Biology and Intercultural Studies (Distinction); Statistics and Health Science.

University of Newcastle Language Centre | Australia

Academic English Program | Jan 2018–Jul 2018

King Abdulaziz University | Saudi Arabia

Preparatory Year Studies (not completed) | 2017–2018

York University English Language Institute (YUELI) | Canada

Academic English Program | 2016–2017

King Fahd University of Petroleum & Minerals (KFUPM) | Saudi Arabia

Preparatory Year Studies (not completed) | 2014–2015